(Dual Space)

Remarks:

① The "zero vector" in $V^* = \mathrm{Hom}(V, \mathbb{F})$
is $\underline{0}_{V^*}: V \to \mathbb{F}$, where $\forall \underline{v} \in V$, $\underline{0}_{V^*}(\underline{v}) = 0$
(为了和$\underline{0}_V$区别)

② In general, $\forall \phi \in \mathrm{Hom}(V, W)$ and $B = \{\underline{v}_i \mid i \in I\}$ basis of V,
$\phi: V \to W$ is uniquely determined
by the values $\phi(\underline{v}_i) \in W$

More explicitly, know the values of $\phi(\underline{v}_i)$ for all $i \in I$
↓
know the value of $\phi(\underline{v})$ $(\forall \underline{v} \in V)$

③ E.g.: For a v. space V with basis $B = \{\underline{v}_i \mid i \in I\}$,
we define $\phi_i \in V^* = \mathrm{Hom}(V, \mathbb{F})$
by $\phi_i(\underline{v}_j) = \delta_{ij}$

e.g.: $V = \mathbb{F}^n$, $B = \{\underline{e}_1, \cdots, \underline{e}_n\}$

Then $\phi_i: \mathbb{F}^n \to \mathbb{F}$ is given by $\phi_i(\underline{e}_i) = 1$, $\phi_i(\underline{e}_j) = 0$, $\forall i \neq j$

$$\Rightarrow \forall \underline{v} = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in \mathbb{F}^n, \quad \phi_i(\underline{v}) = \phi_i(\alpha_1 \underline{e}_1 + \cdots + \alpha_n \underline{e}_n) = \alpha_i$$

$$\Rightarrow \phi_i \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \alpha_i \text{ and } B^* = \{\phi_1, \cdots, \phi_n\}$$

(⇒ 对于上③的 ϕ_i ...=?)

Prop.: $B^* = \{\phi_1, \cdots, \phi_n\}$ is l. indep.

Proof: Take any (∀) finite subset $\{\phi_{i1}, \phi_{i2}, \cdots, \phi_{ik}\} \subseteq B^*$

Consider $\exists \beta_1, \cdots, \beta_k$ not all zeros,

$\beta_1\phi_{i1} + \cdots + \beta_k\phi_{ik} = \underline{0}_{V^*} : V \to \mathbb{F}$ (①)

Evaluate both sides with $\underline{v_j}$, $\underline{\forall j \in [1,k]}$

$\cdots \Rightarrow \forall j, \beta_j = 0 \Rightarrow$ Contradiction

Corollary: If $\dim(V) = n < \infty$,

then B^* is a basis of V^*

Proof: $\dim(V^*) = \dim(\mathrm{Hom}(V, \mathbb{F}))$

(HW4) $= \dim(V) \cdot \dim(\mathbb{F})$

$= \dim(V) \cdot 1 = n$

Since $\dim(V^*) = n$ and $B^* = \{\phi_1, \cdots, \phi_n\}$ is a set of n l. indep. vectors in V^*,

Then B^* must be a basis of V^*

Dual Basis

Def.: If $\dim(V) < \infty$

then B^* is called a dual basis of V^*

Remark: If $\dim(V) = \infty$, then B^* is still l. indep., but not a basis of V^*

E.g.: $V = P(\mathbb{R})$, $B = \{x^i \mid i \in \mathbb{N}\}$

$\Rightarrow B^* = \{\phi_i \mid i \in \mathbb{N}\}$ is l. indep., but $\mathrm{Span}(B^*) \neq V^*$

Consider $\psi \in V^*$ defined by $\psi(p(x)) = p(1)$

Claim: $\psi \notin B^*$

Suppose on contrary

$\psi = r_1\phi_{i_1} + \cdots + r_k\phi_{i_k} \in \mathrm{Span}(B^*)$

W.L.O.G., let $i_1 < i_2 < \cdots < i_k$

Consider $x^{i_k+1} \in V$

Evaluate (注 $L = R$)

$(LHS)\ \psi(x^{i_k+1}) = 1^{x_{k+1}} = 1$

$(RHS)\ r_1\phi_{i_1}(x^{i_k+1}) + \cdots + \sim = 0 + \cdots + 0 = 0$

$\Rightarrow \neq \Rightarrow$ cannot hold

($\Rightarrow V^*$ is "bigger than V in ∞-dim. spaces")

↳ e.g. ($\mathbb{R}^{\mathbb{N}}$ 之一)

Annihilators

Def.: a v. space V, subset $S \subseteq V$

$$\mathrm{Ann}(S) = \{ f \in V^* \mid f(s) = 0, \forall s \in S \} \quad \Rightarrow \mathrm{Ann}(S) \subseteq V^*$$

E.g. $V = M_{2\times 2}(\mathbb{R})$, $B = \{E_{11}, E_{12}, E_{21}, E_{22}\} \subseteq V$

$$B^* = \{\phi_{11}, \phi_{12}, \phi_{21}, \phi_{22}\} \subseteq V^*$$

Let $S = \left\{ \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} \right\} \subseteq V$ (設 $A = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} \in S$)

$\phi_{11}(A) = 1$, $\phi_{12}(A) = 2$,

$\phi_{21}(A) = 0$, $\phi_{22}(A) = -1$

Then $\phi_{21} \in \mathrm{Ann}(S)$

$\phi_{11} + \phi_{22} \in \mathrm{Ann}(S)$

$2\phi_{11} - \phi_{12} \in \mathrm{Ann}(S)$

⋮

Prop, 5.2:

(1) The set $Ann(S)$ is a ~~V~~ subspace of V^*

Proof: Let $\phi_1, \phi_2 \in Ann(S) \subseteq V^*$

(WTS: $\alpha\phi_1 + \beta\phi_2 \in Ann(S)$

Take any $\underline{s} \in S$

$(\alpha\phi_1 + \beta\phi_2)(\underline{s}) = \alpha \cdot 0 + \beta \cdot 0 = 0$

(2) The mapping $Ann(\cdot)$ is inclusion-reversing
↳ i.e.: $\subseteq$

Proof: Take $\phi \in Ann(W_2)$

and consider $\forall \underline{w_1} \in W_1$

Then $\phi(\underline{w_1}) = \phi(\underline{w_1}) = 0$ ($\Leftarrow \phi \in Ann(S)$)

$\in W_1$ $\in W_2$

$\Rightarrow \phi \in Ann(W_1)$

(3) $Ann(S) \overset{=}{(\subseteq)} Ann(Span(S))$

Proof: ① By (2), $S \subseteq Span(S) \Rightarrow Ann(Span(S)) \subseteq Ann(S)$

② (WTS: $Ann(S) \subseteq Ann(Span(S))$

Take $\forall \phi \in Ann(S)$ and $\sum_{i=1}^{k} \beta_i \underline{s_i} \in Span(S)$

$\phi\left(\sum_{i=1}^{k} \beta_i \underline{s_i}\right) = \sum_{i=1}^{k} \beta_i \phi(\underline{s_i}) = \sum_{i=1}^{k} \beta_i \cdot 0 = 0$

$\Rightarrow \phi \in Ann(Span(S))$

(4) $\dim(V) < \infty$, $W \subseteq V$

$\Rightarrow \dim(W) + \dim(\mathrm{Ann}(W)) = \dim(V)$

Proof: Let $\{w_1, \cdots, w_k\}$ be a basis of W

(Basis Extension) $\Rightarrow \{w_1, \cdots, w_k, v_{k+1}, \cdots, v_n\}$ be a basis of V

$\Rightarrow B^* = \{\phi_1, \cdots, \phi_k, \phi_{k+1}, \cdots, \phi_n\}$ be a basis of V^*

Claim: $\mathrm{Ann}(W) = \mathrm{Span}\{\phi_{k+1}, \cdots, \phi_n\}$

If claim holds,

then $\dim(W) + \dim(\mathrm{Ann}(W))$

$= k + (n-k)$

$= n = \dim(V)$

证 claim 成立:

$\forall \beta_{k+1}\phi_{k+1} + \cdots + \beta_n\phi_n \in \mathrm{Span}\{\phi_{k+1}, \cdots, \phi_n\}$,

by dual basis, $\forall 1 \le i \le k$, $(\beta_{k+1}\phi_{k+1} + \cdots + \beta_n\phi_n)(w_i) = 0$

① To check the other inclusion

Take $\forall \phi \in \mathrm{Ann}(\{w_1, \cdots, w_k\})$

Write $\phi = \alpha_1\phi_1 + \cdots + \alpha_n\phi_n$

$\Rightarrow \phi(w_i) = \alpha_1\phi_1(w_i) + \cdots + \alpha_n\phi_n(w_i)$

$0 = \alpha_i$

$\Rightarrow \phi = \alpha_{k+1}\phi_{k+1} + \cdots + \alpha_n\phi_n \in$ ② $\mathrm{Span}\{\phi_{k+1}, \phi_n\}$

E.g.: $V = M_{2\times 2}(\mathbb{R})$,

$S' = \left\{ \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\}$,

$W = Span(S')$

(By (3)) → Then $Ann(S') = Ann(Span(S'))$

$\Rightarrow \dim(\downarrow) = \dim(\downarrow)$

(- - - 忘了)